

Implementation of a CNN Architecture for Pneumonia Detection from Chest X-Ray Images

Student Laboratory Report & Experiment Sheet

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Batch/Division	C	Date	11-8-2026
Faculty	_____	Academic Year	2026-2027

1. Objective

- Understand the workflow of CNN-based image classification.
- Preprocess and load a real-world medical image dataset.
- Design and implement a CNN using TensorFlow/Keras.
- Train a CNN for Normal vs. Pneumonia classification.
- Study the effect of learning rate, batch size and dropout.
- Evaluate the model using accuracy, precision, recall, F1-score and a confusion matrix.

2. Problem Statement

Develop a CNN-based binary image classifier that classifies chest X-ray images as Normal or Pneumonia. This is an educational medical-AI experiment only; the model is not a clinical diagnostic system.

3. Dataset – What Data and Where?

Recommended teaching dataset: “Pneumonia X-Ray Images” on Kaggle. The downloaded dataset contains 5,856 JPEG chest X-ray images in two classes, Normal and Pneumonia. In the uploaded notebook, the observed split was 5,216 training, 16 validation and 624 testing images. The validation folder therefore contains only 16 images, which makes validation accuracy highly sensitive to individual samples.

Dataset: [Kaggle – Pneumonia X-Ray Images](#)

Original/reference dataset: [Kaggle – Chest X-Ray Images \(Pneumonia\)](#)

Folder structure:

dataset/ train/NORMAL,
train/PNEUMONIA
val/NORMAL,
val/PNEUMONIA
test/NORMAL,
test/PNEUMONIA

Observed dataset counts from the uploaded notebook:

Split	NORMAL	PNEUMONIA	Total
Train	1341	3875	5216
Validation	8	8	16
Test	234	390	624
Total	1583	4273	5856

4. Software / Tools

- Google Colab or Jupyter Notebook
- Python 3.x
- TensorFlow / Keras
- NumPy, Matplotlib, Scikit-learn
- GPU runtime recommended

5. CNN Architecture to Implement

Use the following as the fixed baseline architecture. Perform hyperparameter variation only after completing the baseline run.

Layer	Configuration	Purpose
Input	128×128×1	Resize + grayscale
Conv2D-1	32 filters, 3×3, ReLU	Low-level features

MaxPool-1	2×2	Down-sampling
Conv2D-2	64 filters, 3×3, ReLU	Intermediate features
MaxPool-2	2×2	Down-sampling
Conv2D-3	128 filters, 3×3, ReLU	Higher-level features
MaxPool-3	2×2	Down-sampling
Flatten	—	Feature vector
Dense	128, ReLU	Classification representation
Dropout	0.50	Reduce overfitting
Output	1, Sigmoid	Pneumonia probability

Notebook model-summary result: Total parameters = 3,304,193; trainable parameters = 3,304,193; non-trainable parameters = 0.

6. Baseline Hyperparameters

Parameter	Baseline	Action
Image size	128×128 grayscale	Keep fixed
Normalization	Pixel/255 → [0,1]	Keep fixed
Optimizer	Adam	Keep fixed
Learning rate	0.001	Vary
Batch size	32	Vary
Dropout	0.50	Vary
Epochs	15 maximum	Keep fixed / EarlyStopping optional
Loss	Binary cross-entropy	Keep fixed
Output	Sigmoid	Keep fixed

7. Hyperparameters to Vary

Change one major parameter at a time so that the effect of each parameter can be interpreted.

Experiment	Values	Observation
A – Learning rate	0.001, 0.0001	Training stability and validation performance
B – Batch size	16, 32, 64	Training time and validation performance

C – Dropout	0.20, 0.50	Overfitting vs. underfitting
Optional D – Epochs	5, 10, 15	Effect of training duration

Keep the dataset split, preprocessing, CNN architecture, loss function, test set and evaluation procedure fixed for fair comparison.

8. Experimental Procedure

1. Open Google Colab and create a new notebook.
2. Download the dataset from Kaggle and place it in Google Drive or use the Kaggle API.
3. Verify the train/validation/test folders and count images in each class.
4. Display sample Normal and Pneumonia X-rays.
5. Resize images to 128×128 and convert them to grayscale.
6. Normalize pixels to [0,1].
7. Create training, validation and test data loaders.
8. Build the baseline CNN given in Section 5.
9. Compile using Adam, learning rate 0.001, binary cross-entropy and accuracy.
10. Train for a maximum of 15 epochs and monitor validation performance.
11. Plot training/validation accuracy and loss.
12. Evaluate the final model on the unseen test set.
13. Generate a confusion matrix and calculate accuracy, precision, recall and F1-score.
14. Repeat by changing one selected hyperparameter at a time.
15. Record all observations and compare the runs.
16. Select the best configuration and write the conclusion and limitations.

9. Evaluation Metrics

Metric	Meaning
Accuracy	Overall percentage of correctly classified images.
Precision	Of images predicted as Pneumonia, the proportion that are actually Pneumonia.
Recall / Sensitivity	Of actual Pneumonia cases, the proportion correctly detected.
F1-score	Harmonic mean of precision and recall.
Confusion Matrix	Shows TP, TN, FP and FN and the types of errors.

10. Observation – Baseline Run

Run	LR	Batch	Dropout	Epochs	Train Acc.	Val. Acc.	Test Acc.
Baseline	0.001	32	0.50	15	98.88 %	87.50%	74.67%

11. Observation – Hyperparameter Study

- The planned hyperparameter study considered the effect of learning rate, batch size and dropout on CNN performance.
- The learning rate values considered were 0.001 and 0.0001.
- The batch sizes considered were 16, 32 and 64.
- The dropout values considered were 0.20 and 0.50.
- The baseline configuration used a learning rate of 0.001, batch size of 32 and dropout of 0.50.
- The uploaded baseline run did not execute the planned hyperparameter experiments. Therefore, the effect of different learning rates, batch sizes and dropout values cannot be concluded from the current run.
- The hyperparameter study can be performed in future runs to identify the configuration that provides better validation performance and reduces overfitting.

12. Required Outputs / Figures

- Sample images from both classes.
- CNN model summary / architecture.
- Training vs. validation accuracy plot.
- Training vs. validation loss plot.
- Confusion matrix.
- Final classification report.
- Hyperparameter comparison table.
- Optional: ROC curve and AUC.

Note: The uploaded notebook completed the baseline run only. No learning-rate, batch-size, dropout, or epoch variation runs were executed, so no hyperparameter metrics are fabricated.

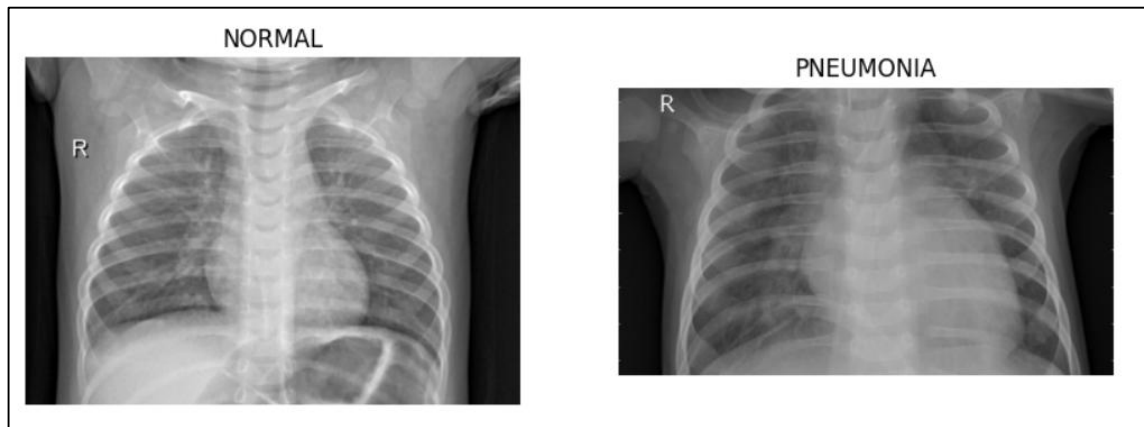


Figure 1. Sample Normal and Pneumonia chest X-ray images generated in the notebook.

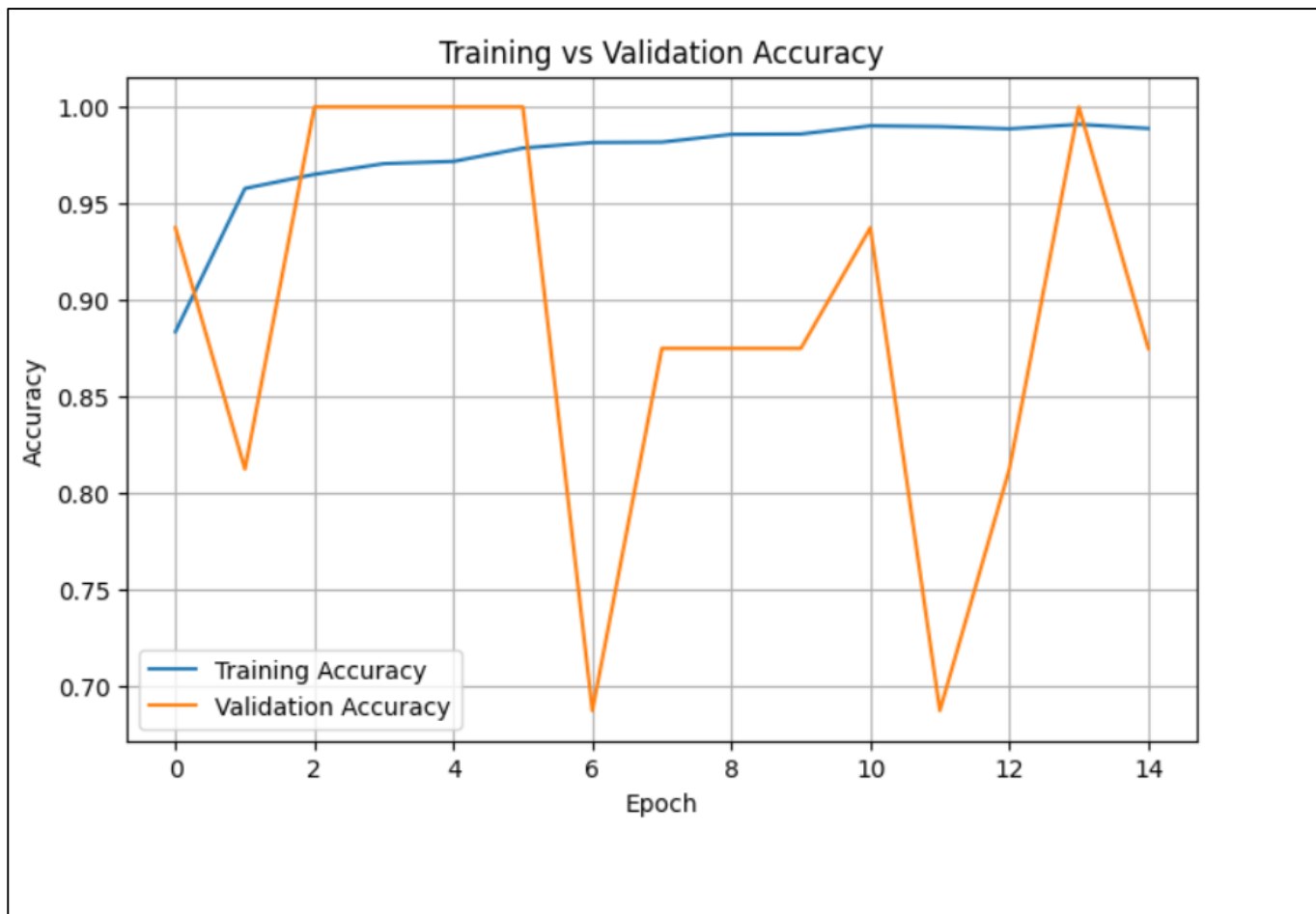


Figure 2. Training and validation accuracy over 15 epochs.

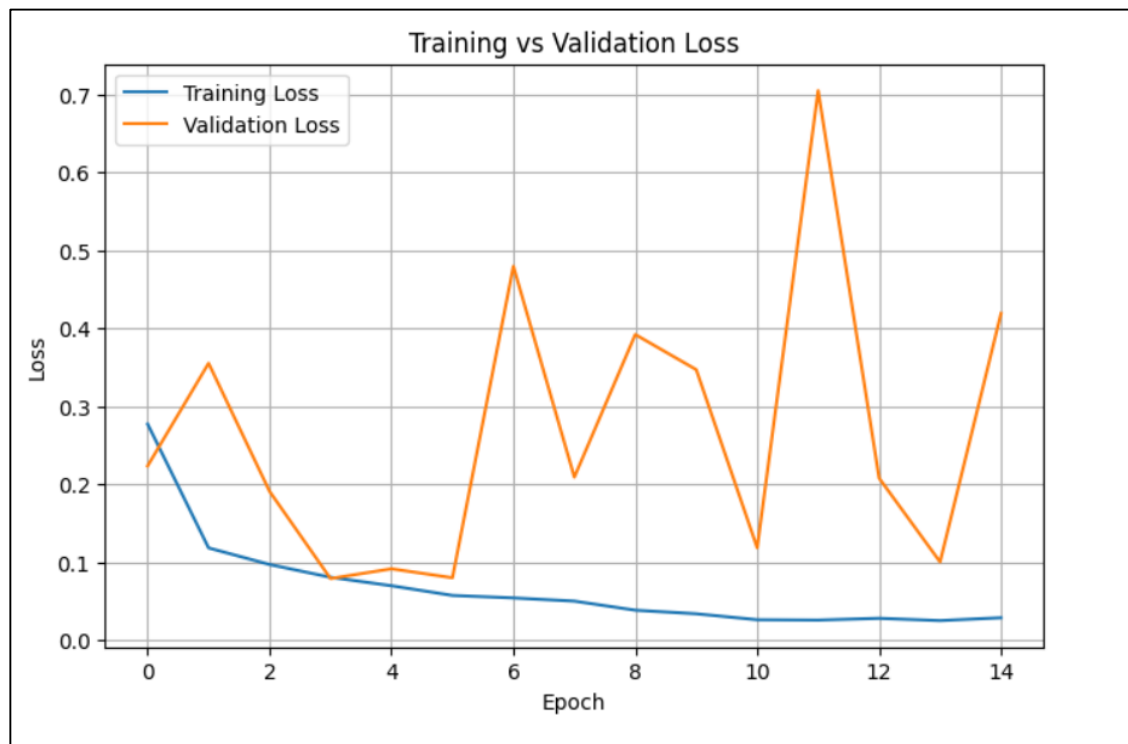


Figure 3. Training and validation loss over 15 epochs.

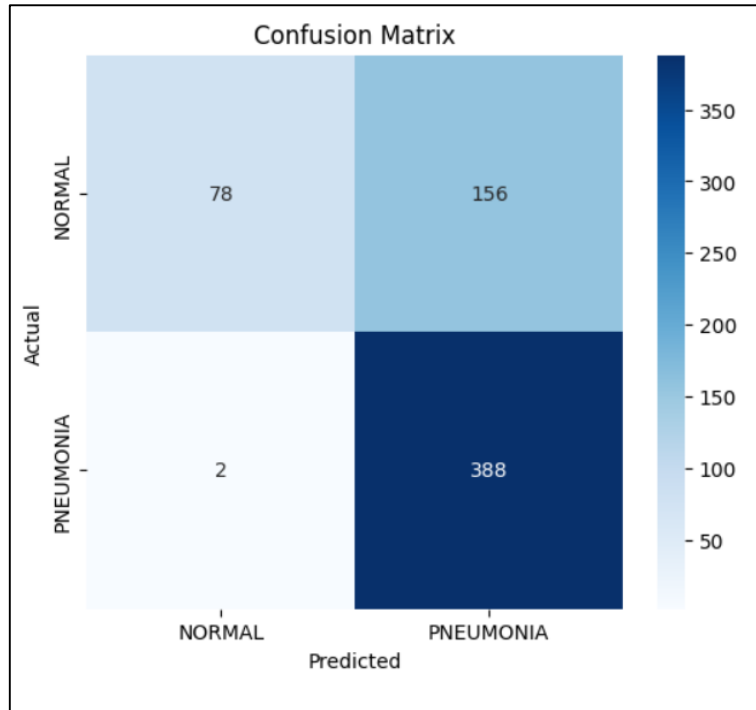


Figure 4. Confusion matrix for the baseline test-set predictions.

Final classification report from the notebook:

Baseline result summary and interpretation

- Final training accuracy: 98.89%; final validation accuracy: 87.50%; test accuracy: 74.68%.
- Test precision for Pneumonia: 71.32%; recall/sensitivity: 99.49%; F1-score: 83.08%.
- Confusion matrix: TN = 78, FP = 156, FN = 2, TP = 388.
- The model detected almost all Pneumonia cases in the test set (high recall), but produced many false positives among Normal images.
- The training accuracy increased to 98.89% while test accuracy was 74.68%, indicating a substantial generalization gap and possible overfitting.
- Validation performance was unstable because the notebook's validation set contained only 16 images. Validation accuracy reached 100% at several epochs but also fell to 68.75% at epochs 7 and 12.
- The lowest observed validation loss was 0.0790 at epoch 4; the final epoch ended with validation loss 0.4190 and validation accuracy 87.50%.

16. Conclusion and Limitations

Conclusion:

A baseline CNN with three convolutional blocks, max-pooling, a 128-unit dense layer, 0.50 dropout and a sigmoid output was successfully implemented using TensorFlow/Keras. The model achieved 74.68% test accuracy, 71.32% Pneumonia precision, 99.49% Pneumonia recall and an F1-score of 83.08%. The very high recall means the baseline missed only two Pneumonia cases in the 624-image test set, but the 156 false positives reduced precision and overall accuracy.

Limitations:

The notebook used only 16 validation images (8 Normal and 8 Pneumonia), so validation accuracy and loss are highly variable and are not a robust basis for selecting a model. The notebook also did not execute the planned hyperparameter study; therefore, the effect of learning rate, batch size and dropout cannot be concluded from the uploaded run. The experiment is educational and must not be treated as a clinical diagnostic system.

Class	Precision	Recall	F1-score	Support
NORMAL	0.98	0.33	0.50	234
PNEUMONIA	0.71	0.99	0.83	390
Accuracy	—	—	0.75	624
Macro avg	0.84	0.66	0.66	624
Weighted avg	0.81	0.75	0.71	624

15. Learning Resources

- [Kaggle – Pneumonia X-Ray Images](#)
- [Kaggle – Chest X-Ray Images \(Pneumonia\)](#)
- [Keras – Image Classification from Scratch](#)

Student Signature: _____ Faculty Signature: _____